

Graphing Quadratics

NAME _____

Complete each section independently. 5-7 minutes total.

Day 1: Vocabulary & Forms

Match each term to its definition:

____ 1. Vertex

____ 2. Axis of Symmetry

____ 3. Standard Form

____ 4. Vertex Form

____ 5. x-intercepts

A. $y = ax^2 + bx + c$

B. The turning point of a parabola

C. $y = a(x - h)^2 + k$ D. Values where $y = 0$

E. Vertical line through the vertex

Quick Question: If a parabola has vertex $(3, -5)$, what is the equation of its axis of symmetry?

Day 2: Perfect Squares Review

Complete the square by filling in the blank, then write as a perfect square:

1. $x^2 + 10x + \underline{\hspace{2cm}} = (x + \underline{\hspace{2cm}})^2$

2. $x^2 - 6x + \underline{\hspace{2cm}} = (x - \underline{\hspace{2cm}})^2$

3. $x^2 + 14x + \underline{\hspace{2cm}} = (x + \underline{\hspace{2cm}})^2$

Challenge: What number completes the square for $x^2 + 5x + \underline{\hspace{2cm}}?$ _____

Day 3: Identifying Features

For the equation $y = -2(x - 4)^2 + 7$, identify:

1. Vertex: _____

2. Axis of Symmetry: _____

3. Opens up or down? _____

4. Wider or narrower than $y = x^2$?

5. Maximum or minimum value? _____

6. What is that max/min value? _____

Day 4: Transformations

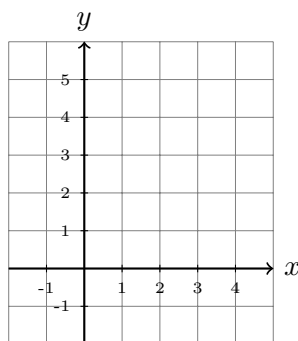
Starting with the parent function $y = x^2$, describe the transformations:

1. $y = (x - 3)^2 + 5$ Shifts: _____

2. $y = -\frac{1}{2}(x + 2)^2$ Transformations: _____

3. $y = 3x^2 - 4$ Transformations: _____

Sketch It: On the grid below, sketch $y = -(x - 1)^2 + 4$ and label the vertex.



Day 5: Converting Forms

Part A: Convert $y = x^2 + 8x + 10$ to vertex form by completing the square.

Vertex form: _____

Part B: Convert $y = (x - 3)^2 - 1$ to standard form by expanding.

Standard form: _____

Quick Check: Circle the form that is easier for finding the vertex: **Standard** or **Vertex**